

# Modeling Reluctance Actuator Topologies with a Focus on Stiffness

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**Abstract**—The reluctance actuator (RA) can provide more acceleration than the Lorentz actuators that are currently in use in next-generation wafer scanners used in semiconductor lithography systems. These wafer scanners are utilized in semiconductor lithography systems. Due to the significant amount of nonlinearity that exists between the RA's input current and output force, it might be challenging to build an effective control system. In the ten years prior, a great number of studies contributed to the modelling, design, and control of RAs; however, the geometrical topology of the actuator was not evaluated from the standpoint of control. This study presents topology selection criteria with the intention of improving the control design process. Additionally, it assesses the various accessible RA topologies, including C and E cores, with reference to the modelling of stiffness. The C-core RA was shown to have a lower magnitude of stiffness at low air gap values (less than 0.5 mm) when compared to the E-core RA. This was discovered through simulation as well as through experimentation.

## I. INTRODUCTION

The next generation of precision motion systems in semiconductor manufacturing machines for the production of Integrated Circuits requires an increase in the actuation system force while maintaining a sufficient level of accuracy [1]. To provide such precision, magnetic actuators are utilized with the most common being the Lorentz actuator [2]. These operate on the Lorentz force and contain a permanent magnet. The primary benefits are a linear force-current relation, and in an ideal range, Lorentz actuators have zero-stiffness where the force is not position dependent [3]. Although they are extensively used currently, the limits on performance in nano positioning systems for Lorentz actuators have been identified [4]. The currently used Lorentz actuator cannot match these requirements, since its size would have to be increased substantially and solving the heat dissipation problem becomes a challenging task [5]. Therefore, the reluctance actuator (RA) has been nominated as a promising candidate for the high precision motion system and particularly for the next generation wafer scanners in the semiconductor lithography systems [6], [7].

RAs work by inducing a magnet field from an electrical current which generates magnetic forces [8], [9]. There however is high nonlinearity between force and current plus

nonlinearities can arise such as hysteresis, fringing, and eddy currents [10]. Some methods to reduce the nonlinearity include PID control [11], observer design [12], and complex hysteresis modeling [13].

The RA was proposed as a part of the high precision positioning system in different studies considering C-core and E-core topologies, see for example [6], [7], [14]. However, there is no standard to select the topology of the RAs. The E-core has a volume core larger than C-core for the same external geometry. This allows the E-core to produce a higher flux density in the air gap compared to C-core for the same input current. On the other hand, C-core may be preferred to easily represent the magnetic circuit and obtain a less error in estimation of the magnetic path length. Such assessment for the topology can be used to estimate the velocity errors and to design a control system that can enhance the performance of the RAs in a class of precision motion applications.

There have been many papers that research control design for photolithography motion systems. For example, a synchronisation control algorithm is created to enhance synchronisation between the reticle and wafer stages in [15]. For very precise motion control systems, a coupled system identification and robust control is demonstrated in [16]. A data-based multi-input multi-output (MIMO) feedforward control method is applied to the motion control of the wafer stage in [17], which has the capacity to mitigate unidentified disturbances in the motion stage. In [18], feedforward control is used to enhance settling time and reduce crosstalk. There has not been any conclusive research yet however on the assessment of stiffness in RA topologies for precision motion.

The paper outlines topology selection criteria to enhance the control design process and evaluates the available RA topologies, including C and E cores, with reference to stiffness. The paper is outlined as follows: an analytical model derivation for the C-core and E-core topologies is presented in Section II. Section III shows the derivation of the stiffness model for both topologies. In Section IV, the model is validated through experimental findings with comparisons to the analytical results and are presented along with a discussion. Finally, Section V features the conclusions of this study and future research is presented.

## II. RELUCTANCE ACTUATOR MODEL

The schematic diagram of the RA is illustrated in [Figure 1 \(a\)](#) for C-core topology with pole width of  $a$ , [Figure 1 \(b\)](#) for C-core topology with pole width of  $2a$  and [Figure 1 \(c\)](#) for E-core topology. The RA comprises a ferromagnetic stator (E-core or C-core) and an I-beam mover.

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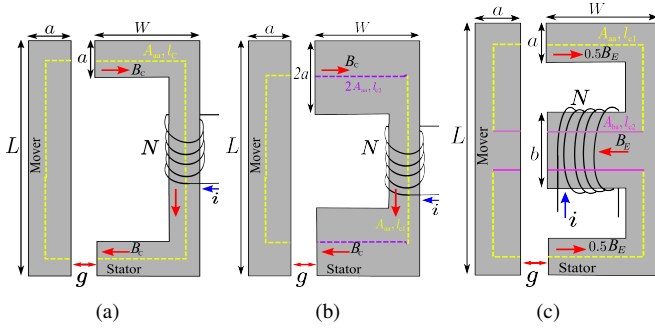


Fig. 1. Schematic diagram of (a) C-Core RA with pole width of  $a$ , (b) C-Core RA with pole width of  $2a$ , and (c) E-Core RA.

The current  $i$  through the coil with  $N$  number of turns generates a magnetic flux  $\Phi$  in the core of the actuators and the air gap  $g$ . Thus, the attractive magnetic force  $F$  is generated in the air gap. In this section, the models of the E-core and C-core RAs with variable air gaps are formulated. The model of the RA is constructed using the magnetic flux  $B(T)$  and the magnetic force  $F$ .

#### A. Model of C-Core RA (Pole width $a$ )

The magnetic model of the electromagnetic actuator is formulated using *Ampere's law* as

$$Ni_C = \oint_l H_C dl, \quad (1)$$

where  $N_C$  is the turns number of the C-core coil,  $H_C$  is the magnetic field in the core,  $l$  is the mean path length (MPL) of the flux path. Which is the average length that the flux will travel through the magnetic core material in one loop [19]. Using the field separation as [20], the magnetic field in the C-core RA is given by

$$H_C = 2H_g + H_f, \quad (2)$$

where  $H_f$  is the magnetic field in the Core ferromagnetic material and  $H_g$  is the magnetic field in the air gaps. Considering the MPL as shown in Figure 1 (a), then *Ampere's law* can be rewritten as

$$Ni_C = 2g \frac{B_C}{\mu_o} + \frac{B_C l_c}{\mu_o \mu_r(B, H)}, \quad (3)$$

where  $B_C$  is the magnetic flux density in the C-core,  $l_c$  is the MPL of the core,  $g$  is the air gap,  $\mu_o$  is the permeability of the free space, and  $\mu_r$  is the relative permeability of the ferromagnetic material based on the BH-curve. Thus, the magnetic flux in the C-core can be expressed by

$$B_C = \psi_C(i_C, g) = \frac{N\mu_o}{\left(\frac{l_c}{\mu_r(B, H)} + 2g\right)} i_C \quad (4)$$

The attractive magnetic force in the air gap of C-core is normal on the I-beam mover and is formulated us-

ing *Maxwell's stress tensor* as [21]

$$F_C = \frac{1}{2\mu_0} \oint_s B_n^2 ds = \frac{1}{2\mu_0} (B_C^2 A_{aa} + B_C^2 A_{aa}),$$

$$F_C = \frac{A_{aa}}{\mu_0} B_C^2 \quad (5)$$

where  $A_{aa} = a^2$  is the cross-section area of the C-core. Thus, the magnetic force of the C-core RA can be expressed as

$$F_C = \frac{\mu_0 A_{aa} N^2}{\left(\frac{l_c}{\mu_r(B, H)} + 2g\right)^2} i_C^2$$

$$F_C = \sigma_C(i_C, g) = \frac{\alpha}{(\beta + 2g)^2} i_C^2, \quad (6)$$

where  $\alpha = \mu_0 A_{aa} N^2$ , and  $\beta = \frac{l_c}{\mu_r(B, H)}$ .

#### B. Model of C-Core RA (Pole width $2a$ )

The structure layout of the C-Core topology of a RA is shown in Figure 1 (b). Following the same steps as outlined previously, the C-core (pole width  $2a$ ) RA can be expressed as

$$F_{C2} = 2 \frac{\mu_0 A_{aa} N^2}{\left(\frac{l_c}{\mu_r(B, H)} + 2g\right)^2} i_C^2$$

$$F_{C2} = \sigma_{C2}(i_C, g) = \frac{2\alpha}{(\gamma + 2g)^2} i_C^2, \quad (7)$$

where  $\gamma = \frac{l_{c1} + l_{c2}}{\mu_r(B, H)}$ .

#### C. Model of E-core RA

Similar to the model of the C-core, field separation [20] can be used to determine the magnetic field to behave by

$$H_{uE} = H_{lE} = H_g + H_{f1} + H_{f2} \quad (8)$$

where  $H_{uE}$  and  $H_{lE}$  are the magnetic fields in the upper and the lower part of the E-core respectively,  $H_{cg}$  is the magnetic field in the center air gap,  $H_{f1}$  is the magnetic field in core with MPL  $l_{c1}$ , and  $H_{f2}$  is the magnetic field in core with MPL  $l_{c2}$ .

Considering the MPLs as shown Figure 1 (b), then the *Ampere's law* can be rewritten as

$$Ni_E = 2 \left( g \frac{B_E}{\mu_o} + \frac{l_{c1}}{2\mu_o \mu_r(B, H)} B_E + \frac{l_{c2}}{2\mu_o \mu_r(B, H)} B_E \right) \quad (9)$$

where  $B_E$  is the magnetic flux density,  $l_{c1}$  and  $l_{c2}$  are the MPLs. Thus, the magnetic flux in the E-core can be expressed by

$$B_E = \psi_E(i_E, g) = \frac{N\mu_o}{\left(\frac{l_{c1} + l_{c2}}{\mu_r(B, H)} + 2g\right)} i_E \quad (10)$$

The attractive magnetic force in the air gap of the E-core is normal on the I-beam mover and is formulated

using *Maxwell's stress tensor* as [21]

$$F_E = \frac{1}{2\mu_0} \oint_s B_n^2 ds = \frac{1}{2\mu_0} \left( 2A_{aa}B_E^2 + \frac{b}{a}A_{aa}B_E^2 \right),$$

$$F_E = \frac{A_{aa}}{\mu_0} \left( \frac{b}{2a} + 1 \right) B_E^2, \quad (11)$$

where  $A_{ab} = \frac{b}{a}A_{aa}$  is the cross-section area of the center lump. Thus, the magnetic force of the RA is given by

$$F_E = \mu_0 A_{aa} N^2 \left( \frac{b}{2a} + 1 \right) \frac{1}{\left( \frac{l_{e1} + l_{e2}}{\mu_r(B, H)} + 2g \right)^2} i_E^2,$$

$$F_E = \sigma_E(i_E, g) = \frac{\alpha \rho}{(\delta + 2g)^2} i_E^2 \quad (12)$$

where  $\rho = \frac{b}{2a} + 1$ , and  $\delta = \frac{l_{e1} + l_{e2}}{\mu_r(B, H)}$ .

The magnetic force formulated by (6), (7) and (12) can be used to derive the formula of the stiffness for the RA, as shown in the next section.

### III. STIFFNESS OF THE RA

According to (6), (7) and (12), the RA exhibits a gap-dependent force, where increasing the air gap correlates to a decrease in the magnetic force. The gap-dependent property of the reluctance contributes to a negative stiffness behaviour, and a restrain guided mechanism such as a compression spring, or flexure structures are necessary for open-loop stability when developing the RA for precision motion system applications [22], [14], [23].

The stiffness of the RA represents the instantaneous change of the magnetic force divided by the instantaneous change of the air gap at the same input current. This section presents the stiffness expressions of the two topologies of the RA.

#### A. Stiffness of the C-core RA

The expression of the stiffness of the C-core RA with pole width of  $a$

$$K_C = \frac{\Delta F_C}{\Delta g}, \quad (13)$$

where  $\Delta F_C$  is the magnetic force's uncertainties due to the air gap  $\Delta g$ .

Now consider, the magnetic force  $F_C^*$  is obtained at input current  $i_C$  and air gap  $g^*$ . Using (6), it can be shown that

$$F_C^* = \frac{\alpha}{(\beta + 2g^*)^2} i_C^2 \quad (14)$$

Assume that  $F_C^* = F_C + \Delta F_C$ , and  $g^* = g + \Delta g$ , then it can be shown that

$$F_C + \Delta F_C = \frac{\alpha}{(\beta + 2(g + \Delta g))^2} i_C^2,$$

$$\Delta F_C = \frac{\alpha}{((\beta + 2g) + 2\Delta g)^2} i_C^2 - \frac{\alpha}{(\beta + 2g)^2} i_C^2,$$

$$\Delta F_C = \left[ \frac{(\beta + 2g)^2}{(\beta + 2g)^2 + 2\Delta g(\beta + 2g) + (2\Delta g)^2} - 1 \right] F_C, \quad (15)$$

thus, the uncertainties in the magnetic force can be represented by

$$\Delta F_C = - \left[ \frac{(2\Delta g)^2 + 2\Delta g(\beta + 2g)}{[(\beta + 2g) + 2\Delta g]^2} \right] F_C \quad (16)$$

Now, the stiffness of the C-core RA (pole width  $a$ ) can be expressed as

$$K_C = - \frac{4\Delta g + 2(\beta + 2g)}{[(\beta + 2\Delta g) + 2g]^2} F_C \quad (17)$$

Similarly, the stiffness of the C-core RA with pole width of  $2a$  can be expressed as

$$K_{C2} = - \frac{4\Delta g + 2(\gamma + 2g)}{[(\gamma + 2\Delta g) + 2g]^2} F_{C2} \quad (18)$$

#### B. Stiffness of the E-core RA

The expression of the stiffness of E-core RA can be given by

$$K_E = \frac{\Delta F_E}{\Delta g}, \quad (19)$$

where  $\Delta F_E$  is the magnetic force's uncertainties due to the air gap  $\Delta g$ .

Now consider, the magnetic force  $F_E^*$  is obtained at input current  $i_E$  and air gap  $g^*$ . Using (12), it can be shown that

$$F_E^* = \frac{\alpha \rho}{(\delta + 2g^*)^2} i_E^2 \quad (20)$$

Assume that  $F_E^* = F_E + \Delta F_E$ , and  $g^* = g + \Delta g$ , then it can be shown that

$$F_E + \Delta F_E = \frac{\alpha \rho}{(\delta + 2(g + \Delta g))^2} i_E^2,$$

$$\Delta F_E = \frac{\alpha \rho}{((\delta + 2g) + 2\Delta g)^2} i_E^2 - \frac{\alpha \rho}{(\delta + 2g)^2} i_E^2,$$

$$\Delta F_E = \left[ \frac{(\delta + 2g)^2}{(\delta + 2g)^2 + 2\Delta g(\delta + 2g) + (2\Delta g)^2} - 1 \right] F_E, \quad (21)$$

thus, the uncertainties in the magnetic force can be represented by

$$\Delta F_E = - \left[ \frac{(2\Delta g)^2 + 2\Delta g(\delta + 2g)}{[(\delta + 2\Delta g) + 2g]^2} \right] F_E, \quad (22)$$

Now, the stiffness of the E-core RA can be expressed as

$$K_E = - \frac{4\Delta g + 2(\delta + 2g)}{[(\delta + 2\Delta g) + 2g]^2} F_E, \quad (23)$$

According to (17), (18), and (23), it can be noted that the stiffness of the RA depends on the magnetic force, the air gap  $g$ , the gap uncertainties  $\Delta g$ , and the actuator geometry represented by  $\beta$ ,  $\gamma$ , and  $\delta$ . Thus, for the same magnetic force, air gap, and gap uncertainties, the C-core RA with pole width  $a$  is expected to have a negative stiffness less than the two other configurations where  $\beta > \gamma$  and  $\beta > \delta$ . Since the other two configurations have larger air gap areas resulting in higher magnetic field strengths,  $H$ , which results in greater  $\mu_r(B, H)$  for these configurations. Also, it can be

TABLE I

THE PARAMETERS OF THE C-CORE AND E-CORE RAS SHOWN IN

FIGURE 1. THESE PARAMETERS WERE CHOSEN BASED ON

CONVENIENCE AND EASE OF MANUFACTURING FOR THE EXPERIMENT.

| Parameter | Value        | Mean Path Length | Value        |
|-----------|--------------|------------------|--------------|
| $N$       | 400          | $A_{aa}$         | $a \times a$ |
| $a$       | 25 mm        | $l_c$            | 350 mm       |
| $b$       | 50 mm        | $l_{c1}$         | 205 mm       |
| $L$       | 140 mm       | $l_{c2}$         | 95 mm        |
| $W$       | 60 mm        | $l_{e1}$         | 150 mm       |
| $A_{ab}$  | $a \times b$ | $l_{e2}$         | 60 mm        |

noticed that for the large air gap, such as the conditions of  $2g \gg \beta$ ,  $2g \gg \gamma$ , and  $2g \gg \delta$  are satisfied, the stiffness of the RA will be the same regardless to the type of the configuration. The next section presents simulation and experimental results to analyze the magnetic actuator stiffness.

#### IV. RESULTS AND DISCUSSION

A comparison has been carried out between C-Core and E-Core RA topologies for the same external geometry as shown in Table I. The used ferromagnetic material is iron-cobalt vanadium soft magnetic alloy *Hiperco 50*. This section presents different simulation results to investigate the stiffness of the C-core and E-core RAs.

##### A. Simulation Results

The magnetic forces using the analytical models (6), (7) and (12) for C-core and E-core RAs are compared for different given air gaps and input currents as shown in Figure 2. It can be noted that the C-core RA with a pole width of  $2a$  can generate a magnetic force equal to the E-core RA, and the two actuators generate a magnetic force larger than that of the C-core RA with a pole width of  $a$ .

To compare the stiffness of the three configurations of the RA, we consider that the three actuators generate the same magnetic force and are affected by the same air gap uncertainties  $\Delta g$ . Figure 3 illustrates the calculated stiffness values using (17), (18), and (23) for a magnetic force of  $F = 100$  N and  $F = 200$  N at a range of the air gaps of  $g = [0.2 : 0.9]$  (mm) and  $g = \{1, 1.2, 1.4, 1.6, 1.8, 2\}$  (mm) with gap uncertainties of  $\Delta g = 10 \mu\text{m}$  around the given air gap  $g$ . This figure indicates that the RA has higher stiffness values with an increase in the magnetic force at specific air gaps  $g$  with gap uncertainties  $\Delta g$ . Also, it can be noticed that at an air gap of  $g > 0.5$  mm, the C-core and E-core RAs almost have the same stiffness values. In contrast, at low air gaps  $g < 0.5$  mm, it can be noted that the C-core RA has a lower stiffness magnitude.

The characteristic of the stiffness for a magnetic force of  $F = 200$  N over a range of air gaps with different gap uncertainties is illustrated in Figure 4. It can be observed that increasing the gap uncertainties leads to an increase in the stiffness value. Also, this figure indicates that the C-core RA with a pole width of  $a$  has a lower stiffness value at a

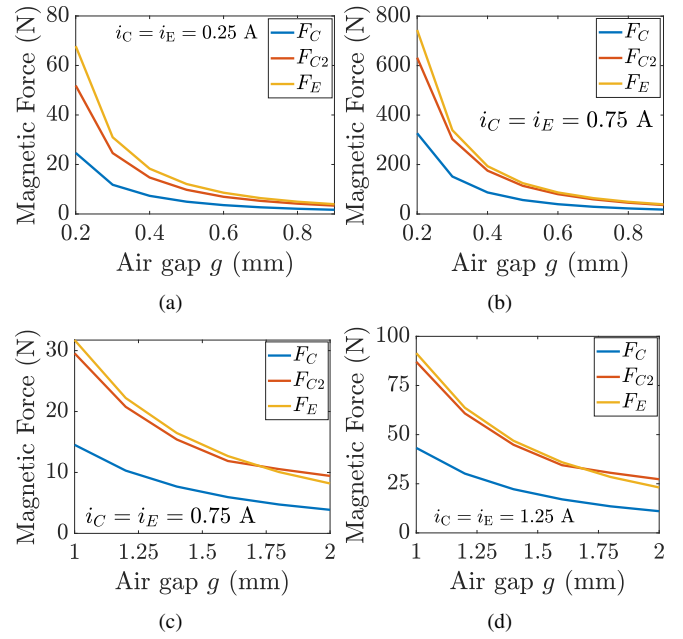


Fig. 2. The comparison between the magnetic force of the RAs under different input current and air gaps of : (a)  $i_C = i_E = 0.25$  A,  $g = [0.2 : 0.9]$  (mm), (b)  $i_C = i_E = 0.75$  A,  $g = [0.2 : 0.9]$  (mm), (c)  $i_C = i_E = 0.75$  A,  $g = \{1, 1.2, 1.4, 1.6, 1.8, 2\}$  (mm), and (d)  $i_C = i_E = 1.25$  A,  $g = \{1, 1.2, 1.4, 1.6, 1.8, 2\}$  (mm).

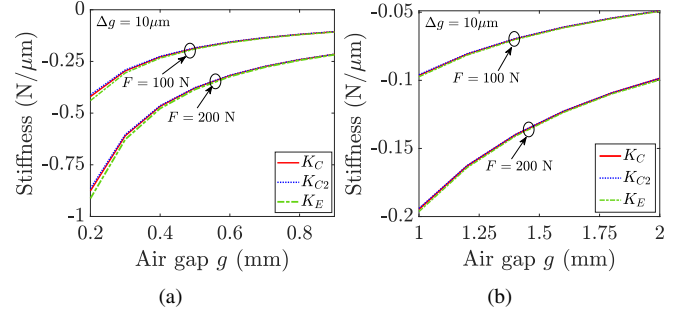


Fig. 3. The comparison between the magnetic stiffness of RAs with different magnetic forces and air gap uncertainties of  $\Delta g = 10 \mu\text{m}$  about the given air gap  $g$  of (a)  $g = [0.2 : 0.9]$  (mm) and (b)  $g = \{1, 1.2, 1.4, 1.6, 1.8, 2\}$  (mm).

small air gap of  $g < 0.5$ , while the C-core and E-core almost have the same stiffness for an air gap of  $g > 0.5$ .

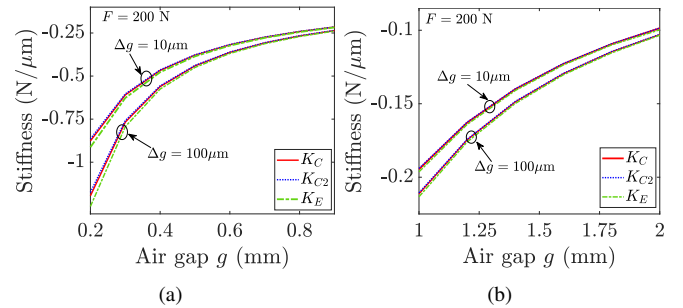


Fig. 4. The comparison between the magnetic stiffness of RAs with different air gap uncertainties and magnetic force of  $F = 200$  N at the given air gap  $g$  of (a)  $g = [0.2 : 0.9]$  (mm) and (b)  $g = \{1, 1.2, 1.4, 1.6, 1.8, 2\}$  (mm).

The simulation results conclude that the stiffness of the RA at given air gaps depends on the magnetic force and the air gap uncertainties regardless of the configuration of the RA for an air gap of  $g > 0.5$  mm. However, for applications that require an air gap of  $g < 0.5$  mm, it is preferred to use a C-core RA with a pole width of  $a$ , which has a lower stiffness.

### B. Experimental Results

The experimental setup is considered to study the characterization of the RA and verify the simulation results under different input currents and air gaps. The schematic diagram of the experimental setup is illustrated in Figure 5. Two enclosures that house the C-core and E-core are made using a 3D printer also used here[19]. The coils of the two actuators are wound on the C-core's two teeth and the E-core's center tooth using American Wire Gauge (AWG) #22 with a maximum current of 7A. The two RAs are driven using a DC power supply with a maximum voltage of 30 V and a maximum current of 5 A. The magnetic force generated in the air gap is measured using four calibrated force sensors. The measured signals are collected using a data acquisition card (dSpace-DS1104) and analyzed using MATLAB software.

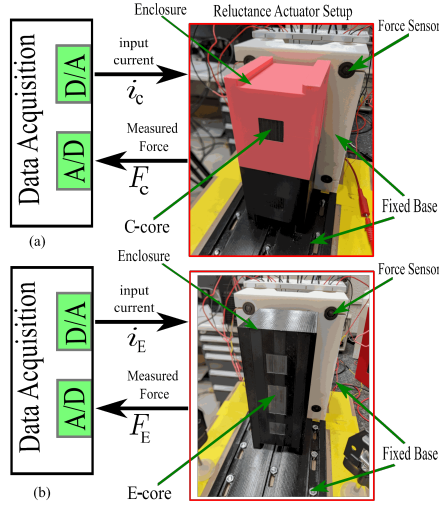


Fig. 5. Schematic diagram of the experimental setup to measure the magnetic force of (a) the C-core and (b) E-core RA. The measured signals are collected using dSPACE Board DS1104.

The measured magnetic forces  $F_C$  for C-core RA at air gaps of  $g = \{1, 1.35, 1.5, 1.65, 2\}$  (mm) and under input currents of  $i_C = i_E = \{0.25, 0.5, 0.75, 1, 1.25, 1.5\}$  (A) are illustrated in Figure 6. While Figure 7 shows the measured magnetic forces  $F_E$  for E-core RA for the same range of air gaps and input currents. It can be noted that the E-core RA generates more force at the same input current and air gap. Figure 8 shows the force as a function of the input current at specific air gaps. The experimental model may present some systematic errors or anomalies in the apparatus as can be observed around the 1.5 mm air gap. The setup can be improved in future experiments.

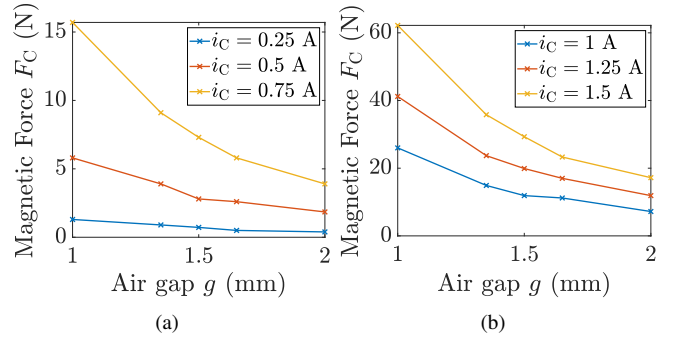


Fig. 6. The measured magnetic force  $F_C$  of C-core RA over air gaps of  $g = \{1, 1.35, 1.5, 1.65, 2\}$  under input currents of: (a)  $i_C = \{0.25, 0.5, 0.75\}$  A and (b)  $i_C = \{1, 1.25, 1.5\}$  A.

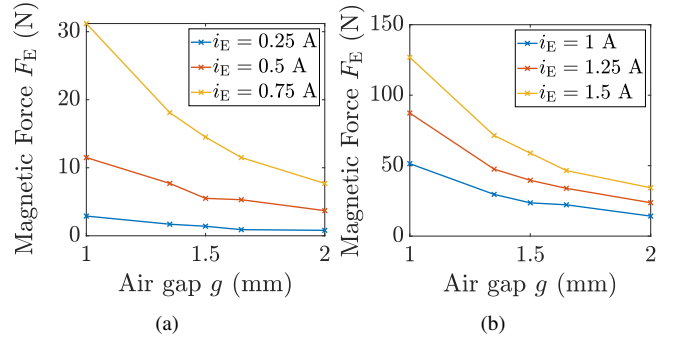


Fig. 7. The measured magnetic force  $F_E$  of E-core RA over air gaps of  $g = \{1, 1.35, 1.5, 1.65, 2\}$  under input currents of: (a)  $i_C = \{0.25, 0.5, 0.75\}$  A and (b)  $i_C = \{1, 1.25, 1.5\}$  A.

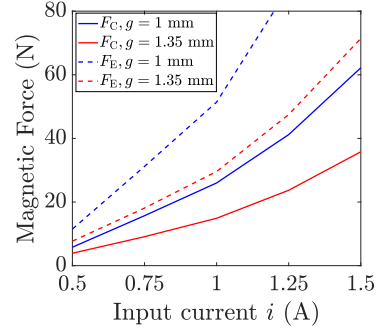


Fig. 8. The comparison between the measured magnetic force  $F_C$  and  $F_E$  at air gap of  $g = 1$  (mm) and  $g = 1.35$  (mm) and the range of input currents considered in the experiments.

Now, these measurements can be used to estimate the magnetic force as a function of the input current and air gap as

$$F_{C_{mes}} \approx \frac{a_1}{g_{mes}^2} i_C^2, \quad (24)$$

$$F_{E_{mes}} \approx \frac{a_2}{g_{mes}^2} i_E^2, \quad (25)$$

where  $i_C$  and  $i_E$  are the input current of the coil,  $g_{mes}$  is the measured air gap, and  $a_1$  and  $a_2$  are coefficients that can be identified using the curve fitting tools as  $a_1 = 2.778 \times 10^{-5}$  and  $a_2 = 5.633 \times 10^{-5}$ .

Now, considering uncertainties  $\Delta g$  around the air gap  $g$  will contribute a force variation  $\Delta F$  that can be formulated



using (24) for the C-core RA as

$$\begin{aligned}\Delta F_C &\approx \frac{a_1}{(g_{mes} + \Delta g)^2} i_C^2 - \frac{a_1}{g_{mes}^2} i_C^2, \\ &\approx \left( \frac{g_{mes}^2}{(g_{mes} + \Delta g)^2} - 1 \right) F_C \approx - \left( \frac{(\Delta g)^2 + 2g_{mes}\Delta g}{(g_{mes} + \Delta g)^2} \right) F_C,\end{aligned}\quad (26)$$

and using (25) for the E-core RA as

$$\begin{aligned}\Delta F_E &\approx \frac{a_2}{(g_{mes} + \Delta g)^2} i_E^2 - \frac{a_2}{g_{mes}^2} i_E^2, \\ &\approx \left( \frac{g_{mes}^2}{(g_{mes} + \Delta g)^2} - 1 \right) F_E \approx - \left( \frac{(\Delta g)^2 + 2g_{mes}\Delta g}{(g_{mes} + \Delta g)^2} \right) F_E,\end{aligned}\quad (27)$$

The stiffness of the C-core RA with pole width of  $a$  can be given by

$$K_C \approx - \left( \frac{\Delta g + 2g_{mes}}{(g_{mes} + \Delta g)^2} \right) F_C, \quad (28)$$

and the E-core RA can be given by

$$K_E \approx - \left( \frac{\Delta g + 2g_{mes}}{(g_{mes} + \Delta g)^2} \right) F_E. \quad (29)$$

Based on the approximated stiffness derived in (28) and (29), it can be noticed that the stiffness value depends on the generated magnetic force at specific air gaps and gap uncertainties. Thus, if the two RAs generate the same amount of magnetic force, then the stiffness is equal. However, with different magnetic force, the C-core RA has a lower stiffness magnitude and therefore can potentially be easier to control.

## V. CONCLUSION

In this paper, the stiffness of the RA is derived as a function of input current and air gap for C-Core and E-Core topologies. The simulation results show that when the air gap displacement is decreased, it leads to an increase in stiffness magnitude. The C-Core RA has a smaller stiffness value compared to the E-Core at the same air gap and the same input current. Selection of a RA topology with a smaller stiffness will decrease the uncertainty in the mover position and improve the error estimation process for the controller. The experimental results are obtained for the C-core RA and E-core RA, and the approximated formulas of the measured magnetic force are used to define the stiffness. It can be concluded that for applications with a motion range of less than 0.5 mm it is preferred to use the C-core RA which has a lower stiffness magnitude, and less force variation with respect to the air gap uncertainties. Future studies will focus on the design and control optimization of the motion system.

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